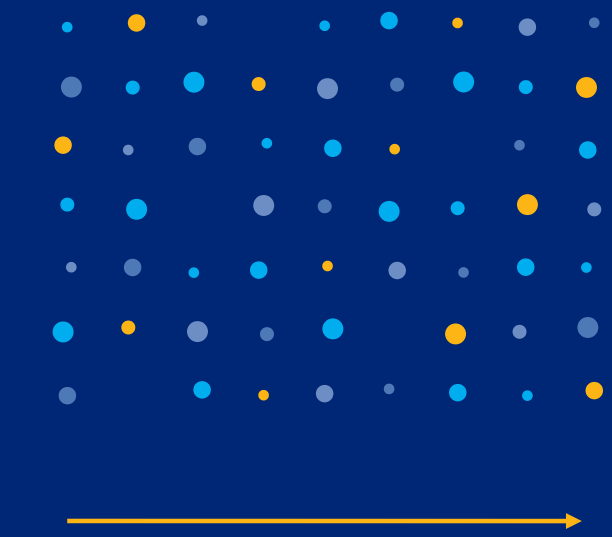
MODEL-ZOO / HEAVY-TAILED SELF-REGULARIZATION

Model weights as evidence

Lineage grouping and performance prediction without descriptive metadata





10,946

lineage models

Phase 2 + ablation

17

weight-derived groups

selected without labels

3.04

modern score error

each model tested once

Completed run: weight_only_2a5abfeb98dc33d9662c

Make predictions first; use descriptive metadata only to check them

Unlabeled discovery

INPUT
complete core + calibrated partial roles

→

FROZEN OUTPUT
clusters + representative models

Known base candidates

INPUT
profiles + fixed candidate list

→

FROZEN OUTPUT
ranked candidate bases

Performance

INPUT
available roles + training scores

→

FROZEN OUTPUT
held-out score estimates

OPEN ONLY AFTER PREDICTION | lineage · family · architecture · model type · parameter count · source · ancestry graph · held-out score

Model IDs only join files. The base-candidate list defines the possible answers, as it does in the reference paper.

Source: weight_only_2a5abfeb98dc33d9662c/metadata.json

Four phases, each ending in a checked output

1 Assemble model lists

Phase 2 and ablation IDs
canonicalize and remove duplicates

19,713 distinct models

2 Analyze weights

read checkpoint matrices
compute alpha by layer and role

10,946 lineage / 11,801 performance

3 Make predictions

form groups · rank known bases
estimate benchmark scores

predictions saved before labels

4 Validate

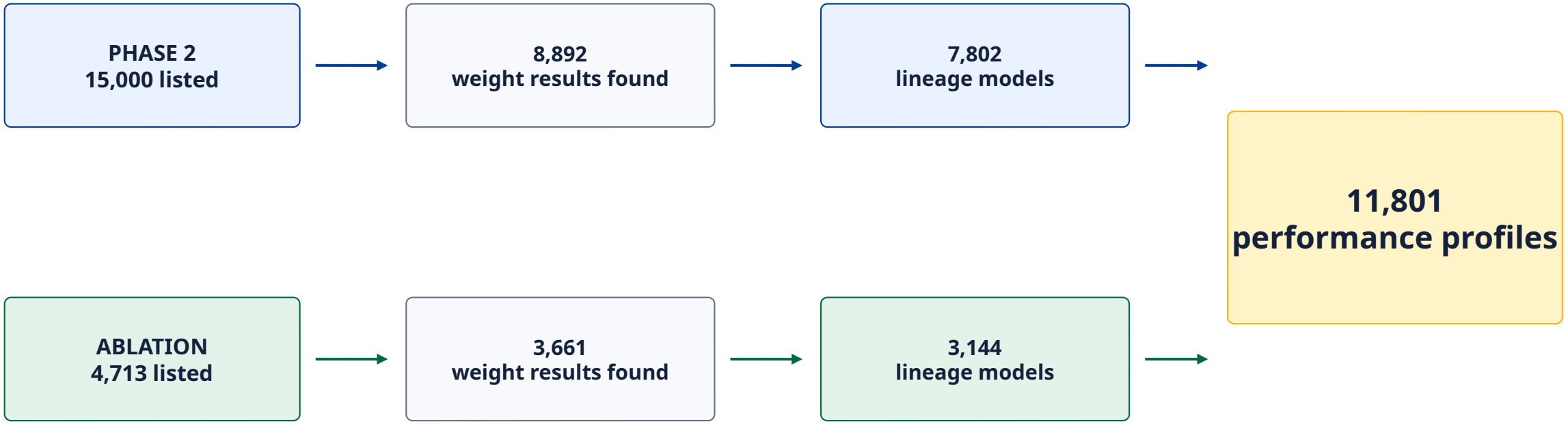
open provenance and held-out scores
measure agreement and error

reported results and limits

The predictor sees model weights, the fixed candidate-base list, and training benchmark scores. Descriptive metadata is opened only in phase 4.

Sources: collection code, completed weight-only run, and post-prediction validation tables.

Shared weight checks lead to task-specific coverage

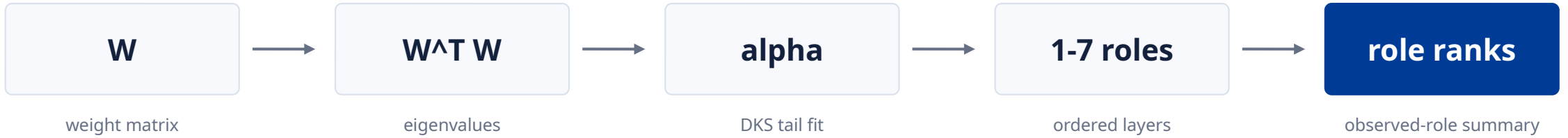


Strict seven-role core: 10,431 models. Reliable missing-role patterns add 515 lineage assignments. Performance uses the label-free cutoff of at least 4 observed roles.

Source labels explain coverage here. They are removed before clustering, candidate-base matching, and performance prediction.

Sources: weight_only_2a5abfeb98dc33d9662c/population_ledger.csv; source-run provenance.json and exclusions.csv

Compare models using the layer roles that are actually observed



Attention: query · key · value · output

Feed-forward: gate · up · down

THIS STUDY

Stored DKS alpha · depth scaled to [0,1] · 32 positions per observed role · within-role ranks · compare only shared roles

REFERENCE PAPER

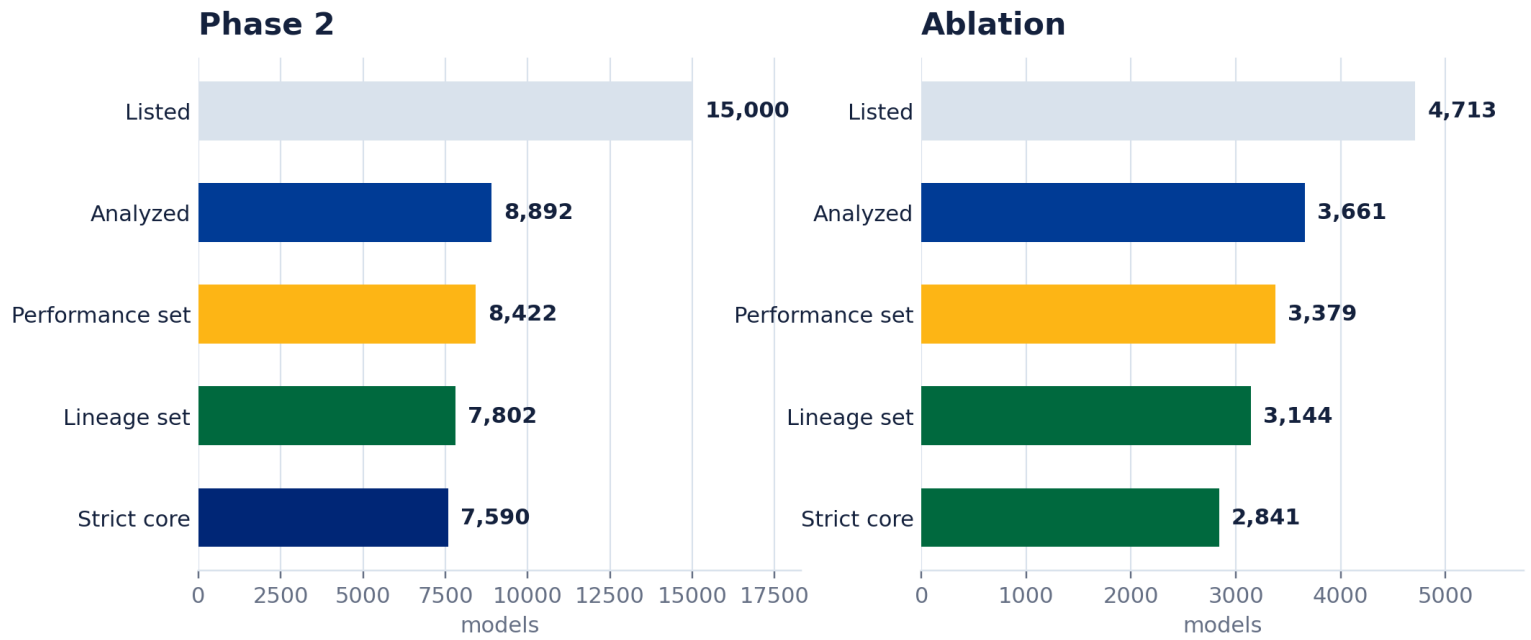
Fixed-tail Hill alpha, k=n/2 · role-wise Spearman · z-scored injective order-preserving dynamic programming when depths differ

Same scientific idea; different alpha estimator and unequal-depth calculation. Numerical results are not an exact replication.

Sources: stored DKS alpha values by layer; archive/reference-work/Spectral Signatures of Large Language Models.html

Seven roles define the paper-comparison core, not general eligibility

Strict paper comparison and broader operational coverage



10,431

strict seven-role core

10,946

lineage assignments

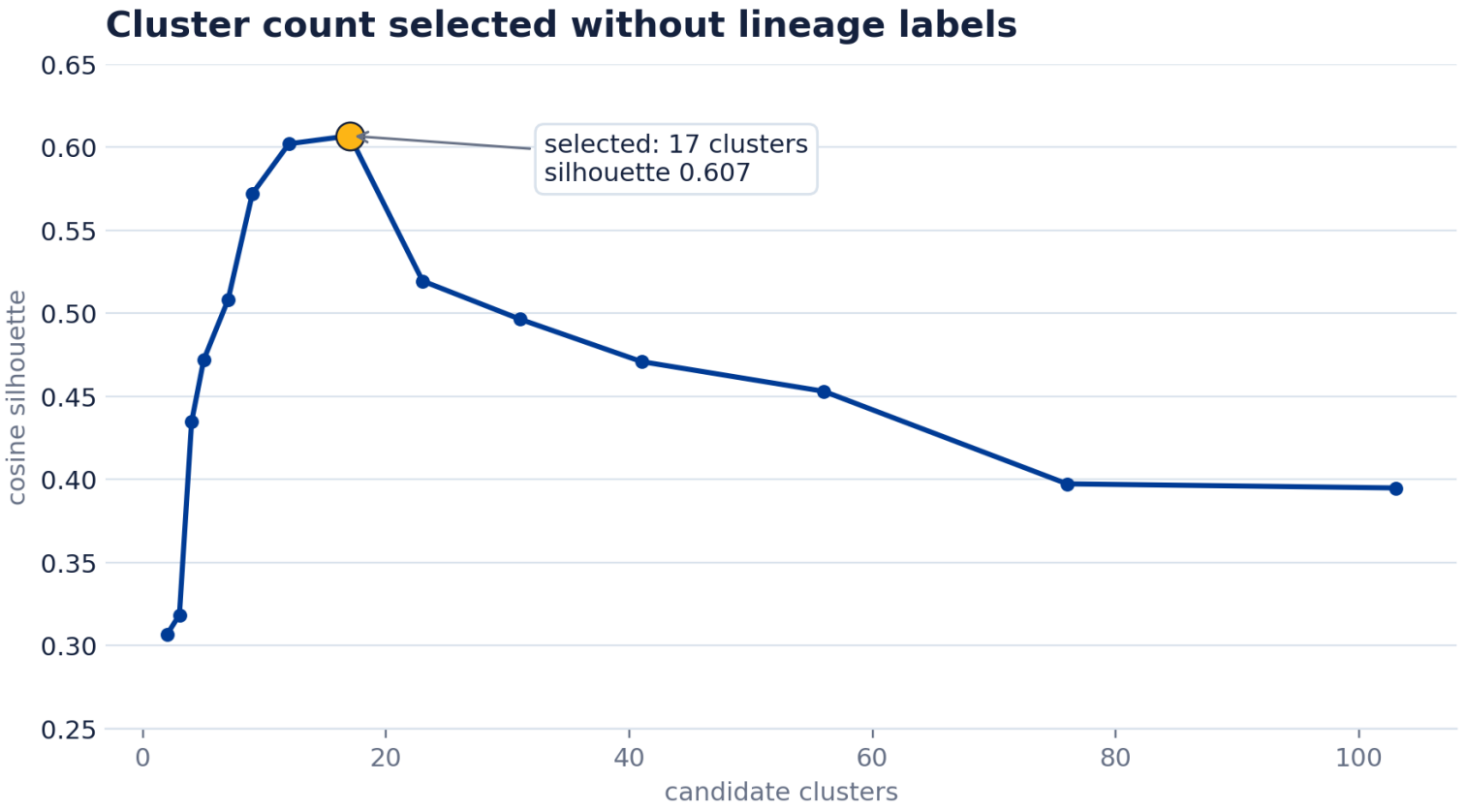
11,801

performance profiles

Partial lineage patterns must retain the complete-profile nearest model in the top ten at least 90% of the time. Performance accepts the label-free 4-role cutoff and requires every neighbor to share that many roles.

Sources: weight_only_2a5abfeb98dc33d9662c/population_ledger.csv and metadata.json

17 core groups give the clearest separation without family labels



0.607

group separation

higher is better

0.944

repeat-run agreement

average of 3 reruns

0.883

lowest rerun agreement

1.541

within/between spread

lower is better

HOW 17 WAS CHOSEN

Fit candidate counts on 10,431 complete profiles

Keep the count with the best separation

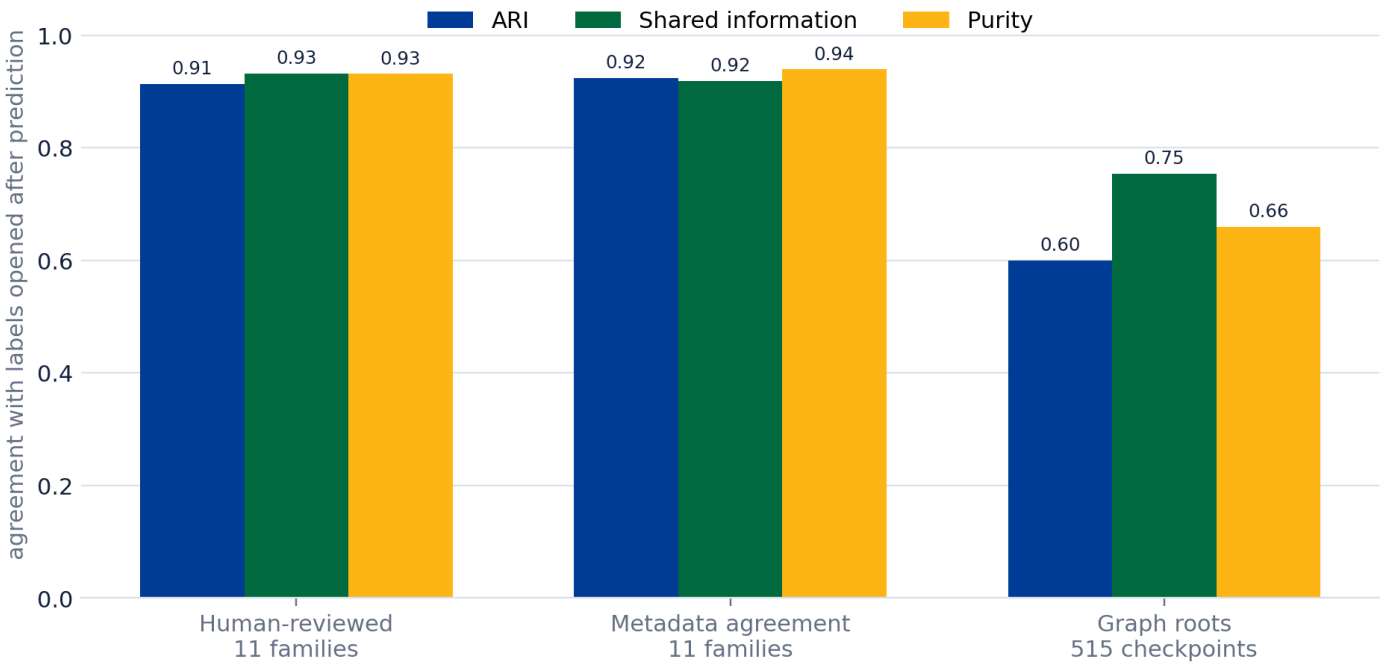
515 partial profiles are attached only after label-free masking checks pass.

Sources: weight_only_2a5abfeb98dc33d9662c/lineage_cluster_selection.csv and lineage_internal_metrics.csv

VALIDATION AFTER PREDICTION

Compare the 17 weight-derived groups with three independent label sets

Broad families align best with the 17 discovered groups



LABELS OPENED AFTER GROUPING

Human-reviewed
174 models assigned to 11 broad base families by manual review.

Metadata agreement
2,981 models assigned to 11 broad families when metadata rules agree.

Graph roots
5,013 models traced to 529 individual root checkpoints in the provenance graph.

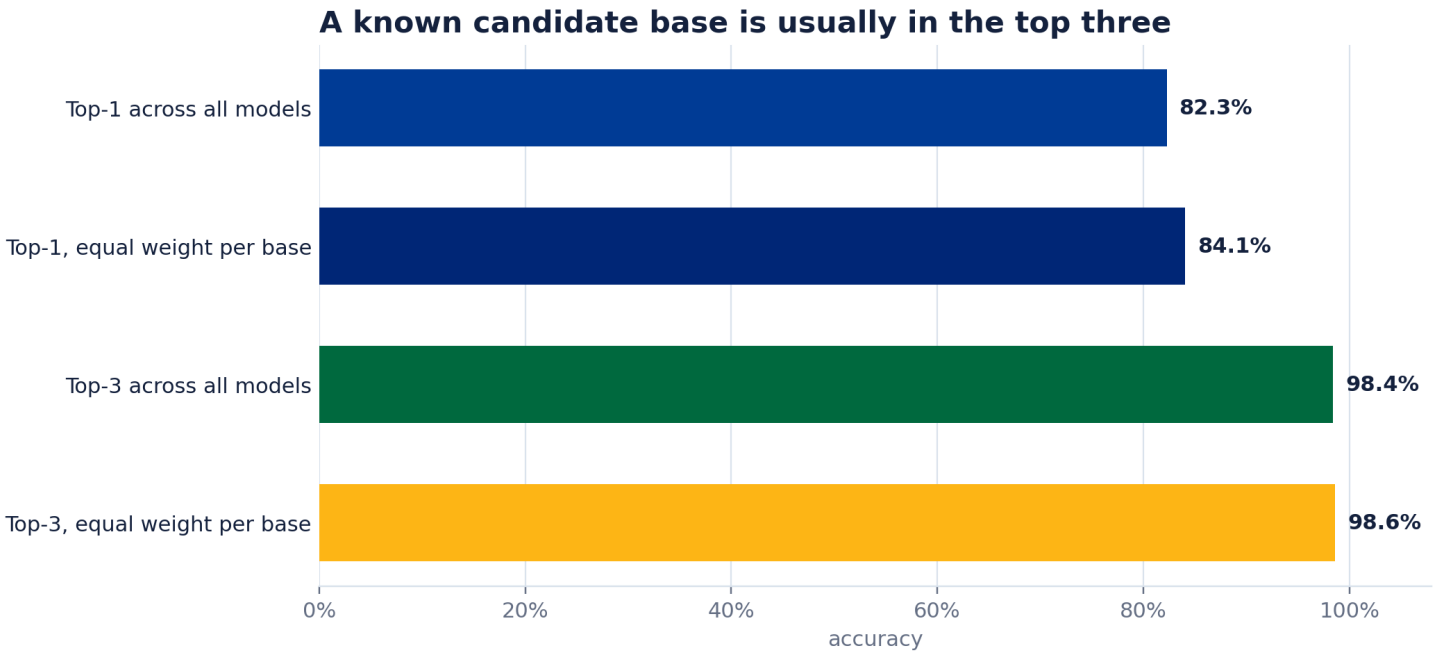
ARI
Pairwise grouping agreement, corrected for chance. 1 means identical grouping.

Shared information
How much knowing one grouping tells us about the other. 1 is complete.

Purity
Majority-label share inside each predicted group. It can favor too many small groups.

Sources: weight_only_2a5abfeb98dc33d9662c/lineage_validation_metrics.csv and lineage_cluster_assignments.csv

Match each model against a fixed list of known base checkpoints



29 / 38

available / supplied bases

3,836

models scored

82.27%

top choice correct

across all models

84.07%

top choice correct

each base weighted equally

The possible answers are fixed before prediction. A model is compared with all 29 available base profiles and the bases are ranked by similarity.

A base outside this candidate list cannot be selected. Provenance is opened afterward only to score the ranking.

Sources: weight_only_2a5abfeb98dc33d9662c/lineage_gallery_metrics.csv; reference paper Sec. 5.2 and 5.4

The evaluation boundary matches; the estimator and scale do not

Dimension	Reference paper	Current study
Model selection	Public provenance	Fixed candidate list + labels opened afterward
Queries / bases	128 / 8	3,836 / 29 profiled
Prediction input	Weights only	Weights only
Alpha	Fixed-tail Hill	Stored DKS
Unequal depth	Z-score + order-preserving dynamic programming	32-point normalized-depth interpolation
Overall top-1	98.44%	82.27%

Do not interpret the accuracy gap as an estimator ranking: the candidate sets, labels, coverage, alpha fit, and depth handling differ.

Paper: arXiv:2607.03377v1. Current metrics: lineage_gallery_metrics.csv.

Every scored model is tested using models outside its test part

BUILD FIVE TEST PARTS

Group identical observed-role summaries → Divide those groups into five parts → Save the division before prediction

REPEAT FIVE TIMES

Hold out 1 part → Use the other 4 parts → Find 3 closest training models → Predict a weighted score average → Open the held-out score and measure error

A fold is one of the five parts. Each part takes one turn as the held-out test set.

Identical summaries stay together. Closer spectral matches receive more weight.

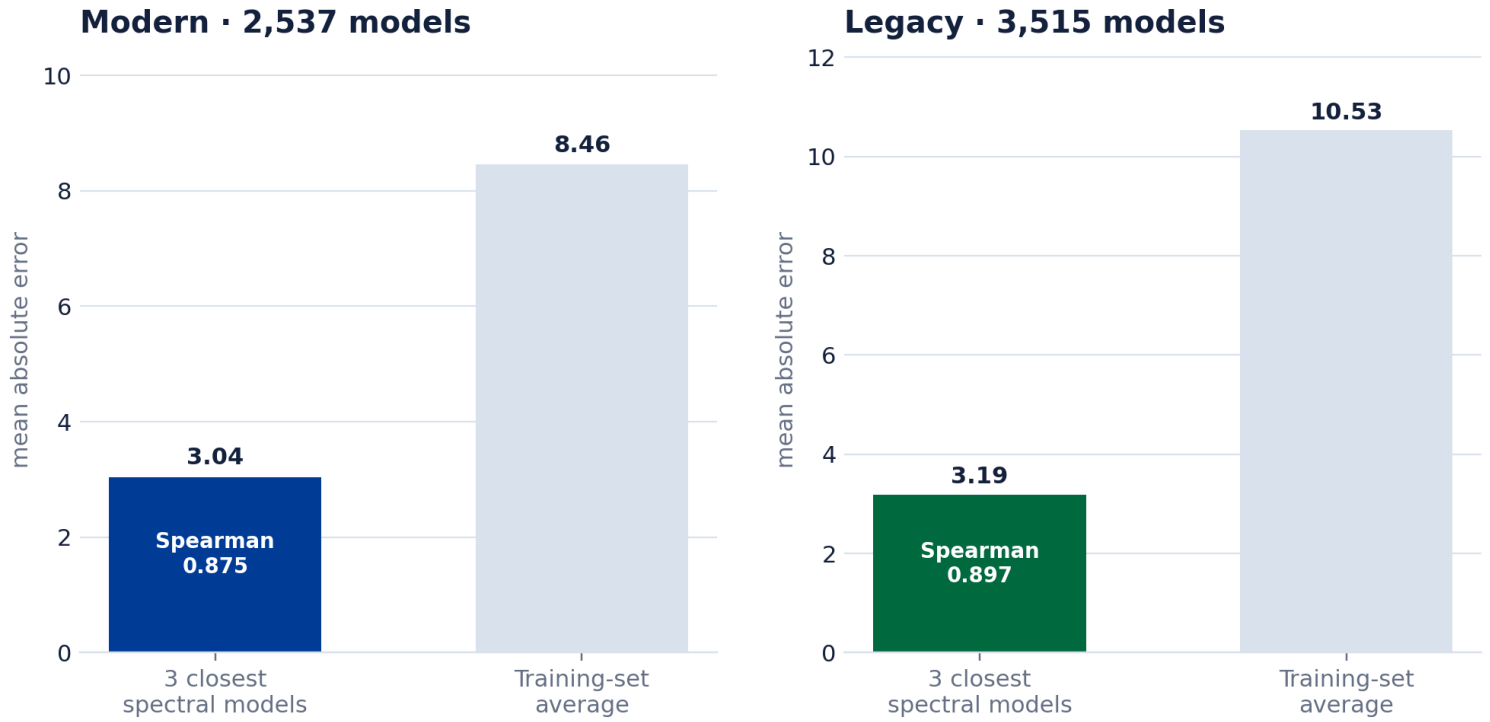
Modern and legacy benchmark sets are trained and evaluated separately. Descriptive metadata is never a predictor input.

Sources: weight_only_2a5abfeb98dc33d9662c/performance_frozen_splits.csv and performance_prediction_freeze.csv

PERFORMANCE RESULTS

Nearest spectral neighbors predict well when similar models are represented

Each model is tested using the other four data parts



Modern

3.043 MAE

0.875 Spearman

2,537 models / 2,410 unique spectral summaries

Legacy

3.188 MAE

0.897 Spearman

3,515 models / 3,363 unique spectral summaries

These results cover models with similar weight summaries in training. The next test removes complete weight groups.

Source: weight_only_2a5abfeb98dc33d9662c/performance_metrics.csv

Holding out a model's entire weight group reveals the limit

Prediction is harder when a complete weight-derived group is absent



Modern predictions: ordinary test versus a held-out weight group



MODERN TASKS
MAE 7.721 · Spearman 0.479
training average 9.102

LEGACY TASKS
MAE 14.169 · Spearman 0.032
training average 13.182

For legacy tasks, this is worse than using the training-set average. Similarity must be checked before trusting a score estimate.

Sources: weight_only_2a5abfeb98dc33d9662c/performance_cluster_holdout_metrics.csv and performance_cluster_holdout_predictions.csv

What the results show, and where uncertainty remains

Observed	Interpretation	Remaining uncertainty
17 repeatable weight-derived groups	Alpha order through depth contains family information	Similarity cannot establish parent or training direction
Known base appears in top 3 for 98.4%	Broad families are easier than exact checkpoint ancestry	A base absent from the candidate list cannot be returned
Modern-score error is 3.043 in the ordinary test	Score estimates depend on nearby training examples	Legacy errors exceed the mean baseline when a whole group is unseen

Overall: weight-derived alpha summaries support broad grouping, known-base matching, and local score estimation. Exact ancestry needs an additional direct weight-matrix check.

Summary of the frozen lineage and performance evaluations in the completed run.

APPENDIX

Methods and definitions

Operational details, next weight-based methods, and terms used in the main results

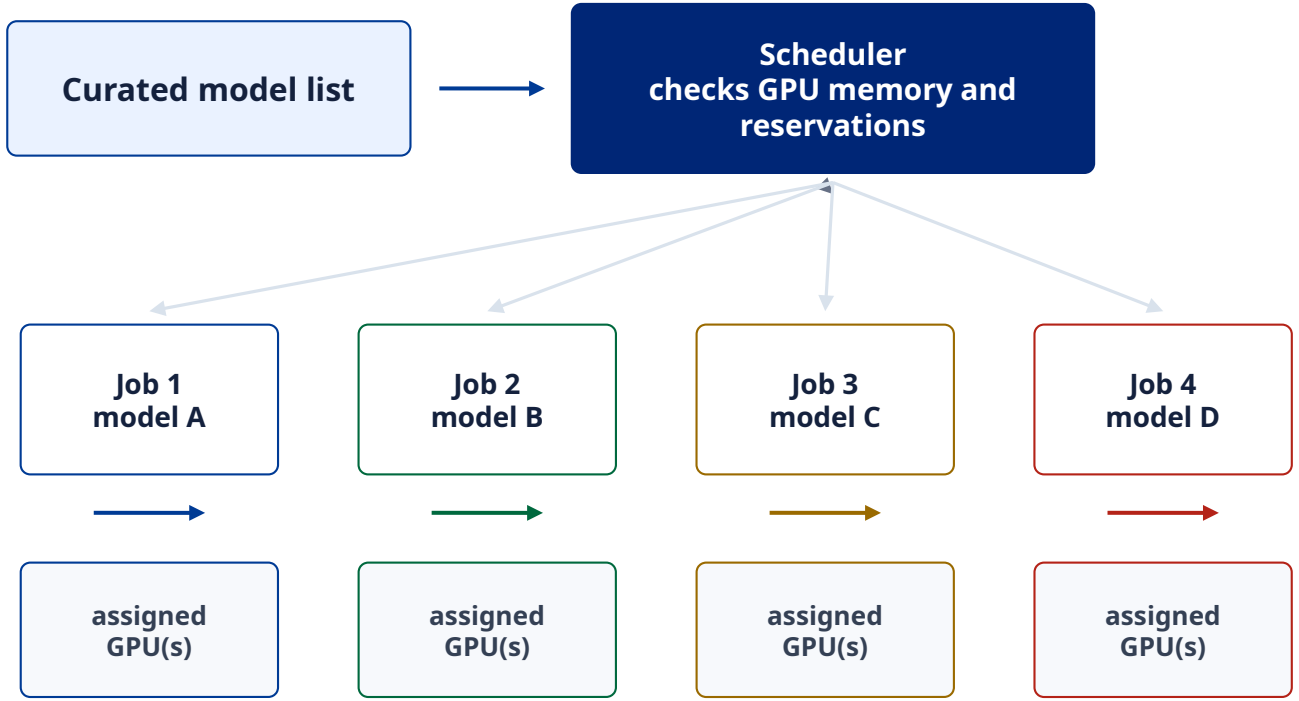
A1 — **Collection and GPU scheduling**

A2 — **Matrix analysis and storage**

A3 — **Future direct weight comparison**

A4 — **Terms and evidence guide**

One scheduler assigns jobs to available graphics processors



CONFIGURABLE CONTROLS

- which GPU IDs may be used
- GPUs per job and jobs at once
- memory threshold before assignment
- retry count and stale-job timeout
- reload, drain, or stop while running

Each job uses its own download cache. Removing it after the job prevents storage growth, at the cost of downloading shared files again.

The memory check and internal reservation avoid overlap among these jobs; they are not a cluster-wide lock against unrelated users.

Sources: Model-Zoo/esd_experiment/gputracker and Model-Zoo/docs/operations/analysis.md

Each weight matrix becomes a fitted spectrum and a structured record



PLUG-IN ANALYSIS CHOICES

DKS
choose the tail start that best fits

Fixed rules
use a specified part of the tail

STRUCTURED OUTPUTS

CSV / HDF5
per-layer metrics and model fields

Optional eigenvalues
larger files; useful for later re-fitting

Logs and status
progress, failures, retries, and timing

Matrix work can be split across several GPUs. This lowers wall-clock time when matrix operations dominate, but does not guarantee fewer total GPU-seconds.

Analysis continues past inaccessible or repeatedly failing models after the configured retry limit. Completed records remain available for aggregation.

Sources: Model-Zoo/esd_experiment/src/worker.py and Model-Zoo/README.md

Add direct weight-matrix comparisons for within-family verification

STAGE 1
HTSR coarse retrieval
10,946 assigned models

→

STAGE 2
direct weight-matrix check
small candidate set

→

OUTPUT
related / unrelated evidence
with calibrated thresholds

arXiv:2608.07786

Spectral energy + subspace alignment

Coarse family separation plus fine directional geometry.

arXiv:2608.14929

Centered residual signatures

Removes shared identity structure before comparing residual blocks.

AWM · arXiv:2510.06738

Assignment + centered alignment

Aligns matrices, then compares their centered weight structure.

Not tested here: the saved cache has scalar eigenvalue summaries, not singular vectors or centered weight matrices.

Primary sources: arXiv:2608.07786, arXiv:2608.14929, arXiv:2510.06738.

Measures used in this study

Alpha — Exponent fitted to the upper eigenvalue distribution of a weight matrix.

DKS — Chooses the tail start that minimizes the Kolmogorov-Smirnov fit distance.

ARI — Adjusted Rand Index: grouping agreement corrected for chance; 1 is exact.

Spearman — Agreement in rank order; 1 means the two rankings match.

MAE — Mean absolute error: the average size of a score prediction error.

Five-part test — Divide models into five fixed parts; test each part using the other four.

Known-base test — Choose the closest model from a fixed list of possible base checkpoints.

Representative — The observed core model closest to its group mean; not necessarily its historical base.

Evidence boundary: model weights and benchmark training scores create predictions. Provenance and family metadata are opened only after predictions are saved, to measure correctness.

Definitions follow the analysis code and cited reference methods.